Google Cloud

Course 2 resources and citations

Module 1: Introduction to data management and storage in the cloud

Resources

Helpful resources and tips

- Resolve issues with a lab
- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

Guide to BigQuery

- Enable the BigQuery sandbox
- Load and query data

Citations

Data storage and connections

- Google Cloud. (n.d.). *Cloud storage*. Retrieved December 1, 2023.
- Google Cloud. (n.d.). *What are transactional databases?* Retrieved April 24, 2023.
- Google Cloud. (2023, November 30). *Product overview of Cloud Storage*. Retrieved November 30, 2023.
- Indeed editorial team. (2022, June 24). *Systems of record: What they are, common types and benefits*. Indeed.

Common ways to store data

- Google Cloud. (n.d.). *Google Cloud databases*. Retrieved December 1, 2023.
- Google Cloud. (n.d.). *What is a data lake?* Retrieved December 1, 2023.
- Google Cloud. (n.d.). *What is a data warehouse?* Retrieved December 1, 2023.
- Joubert, S. (2016, May 13). How much data is produced every day? *Graduate Blog*.

[Supplemental] Common data storage systems

- Google Cloud. (n.d.). *What is NoSQL?* Retrieved November 16, 2023.
- Google Cloud. (n.d.). *What is a relational database?* Retrieved November 16, 2023.
- Madison, M., Barnhill, M., Napier, C., & Godin, J. (2015). NoSQL database technologies. *Journal of International Technology and Information Management, 24(1)*.

Structured, unstructured, and semi-structured data

- Coursera. (2023, November 29). *SQL vs. NoSQL: The differences explained + when to use each*.
- Google Cloud. (n.d.). *BigQuery documentation*. Retrieved April 25, 2023.
- Google Cloud. (n.d.). *Cloud storage*. Retrieved April 24, 2023.
- Google Cloud. (2020, March 7). *Structured and unstructured storage in the cloud* [Video]. Youtube.

Overview of data lakehouse architecture

- Google Cloud. (n.d.). *What is a data lakehouse?* Retrieved December 1, 2023.
- Kortvelyesi, P. (2017, January 7). Top 7 challenges of building a data lake. *Dataprd.Com*.

Example of a data lakehouse

- Broadfoot, K. (2021). *A practical guide to cloud migration*. O'Rielly.
- Google Cloud. (n.d.). *What is a data lakehouse?* Retrieved December 1, 2023.
- Google Cloud. (2023, November 29). *Specifying a schema*. Retrieved November 30, 2023.

Aspects of table schema

- Google Cloud. (2023, November 29). *Data types*. Retrieved November 30, 2023.
- Google Cloud. (2023, November 29). *Modifying table schemas*. Retrieved November 30, 2023.
- Google Cloud. (2023, November 29). *Schema design best practices*. Retrieved November 30, 2023.
- Google Cloud. (2023, November 29). *Specifying a schema*. Retrieved November 30, 2023.

Components of BigQuery table schema

- Google Cloud. (2024, January 2). *Specifying a schema*.

Introduction to nested data structure

- Google Cloud. (2023, November 29). *Use nested and repeated fields*.
- Stanford University. (n.d.). [Lecture notes on *Nested data structures*]. Department of Computer Science, Stanford University.

Guide to BigQuery

- Google Cloud. (January 2, 2024) *Enable the BigQuery Sandbox*.
- Google Cloud. (January 2, 2024) *Google Cloud overview*.
- Google Cloud. (January 2, 2024) *Organizing BigQuery resources*.

- Google Cloud. (January 2, 2024) *Query a public dataset with the Google Cloud console*.

Overview of data processing methods

- Apache NiFi. (2023, November 23). *Apache NiFi overview*.
- Google Cloud. (2023, November 30). *Loading CSV data from Cloud Storage*. Retrieved December 1, 2023.
- JSON. (n.d.). *Introducing JSON*.
- Kafka. (n.d.). *Kafka 3.6 documentation*.

Batch versus streaming data processing

- Linkedin. (2023, November 16). *What are the differences between batch and streaming cloud pipelines?* Linkedin.
- Walker, M. (n.d.). Batch vs. real time data processing. *Michael Walker's Blog*.

Module 2: Key components of data organization

Citations

Denormalized data

- Department_of_C. (2021, June 14). *Database (Concept): Data normalization: 1st, 3rd normal form*. Looker.
- Thallam, R. (2020, September 30). BigQuery explained: Working with joins, nested & repeated data. *Google Cloud Blog*.

Data governance for effective data management

- Google Cloud. (n.d.). *What is data governance?* Retrieved December 1, 2023.
- Qureshi, I. (2022, March 14). Data governance in the cloud—Part 2—Tools. *Google Cloud Blog*.

Introduction to master data management

- Deighton, A., & Arsanjani, Dr. A. (2023, January 20). Built with BigQuery: How Tamr delivers master data management at scale and what this means for a data product strategy. *Google Cloud Blog*.
- Ichhpurani, K., & Hasbe, S. (2019, May 21). Delivering end-to-end data analytics and data management solutions with Informatica. *Google Cloud Blog*.

Introduction to data catalogs

- Google Cloud. (2023, November 29). *About data lineage*. Retrieved December 1, 2023.
- Linkedin. (n.d.). *What are the key features and functionalities of a data catalog that you look for?*

Technical and business metadata

- Bapat, S. (2020, April 30). Discover, understand and manage your data with Data Catalog, now GA. *Google Cloud Blog*.
- Dennis, A. L. (2022, July 5). *Demystifying metadata management for tech and business*. DATAVERSITY.
- Google Cloud. (2023, December 20). *Data catalog metadata*.

Overview of data lakehouse architecture

- Google Cloud. (n.d.). *What is a data lakehouse?* Retrieved December 1, 2023.
- Levy, R., Thill, S., & Tekiner, F. (2021). *Building a data lakehouse on Google Cloud Platform*. Google Cloud.
- Tekiner, F., & Pierce, S. (2021, October 28). Open data lakehouse on Google Cloud. *Google Cloud Blog*.

Components of data lakehouse architecture

- Google Cloud. (n.d.). *What is a data lakehouse?* Retrieved December 1, 2023.
- Levy, R., Thill, S., & Tekiner, F. (2021). *Building a data lakehouse on Google Cloud Platform*. Google Cloud.
- Tekiner, F., & Pierce, S. (2021, October 28). Open data lakehouse on Google Cloud. *Google Cloud Blog*.

Module 3: Steps to find data

Resources

How to find data using BigQuery

- BigQuery explained: How to query your data
- BigQuery public datasets

Dataplex's data lineage feature

- Understand and trust data with Dataplex data lineage

How to use the Dataplex data lineage feature

- Use data lineage with Google Cloud systems

Analytics Hub enables data sharing

- Introduction analytics hub

How to use Analytics Hub

- Manage data exchanges
- Manage subscriptions

Overview of Dataplex

- Dataplex overview
- Preview: Google Cloud Dataplex wows

Citations

How to find data using BigQuery

- Google Cloud. (2024, January 18). *BigQuery public datasets*.

Data lineage and traceability

- Google Cloud. (2023, November 29). *About data lineage*. Retrieved December 1, 2023.
- Looker Studio Help. (n.d.). *About data sources*. Retrieved December 1, 2023.

Dataplex's data lineage feature

- Google Cloud. (2024, January 2). *About data lineage*.
- Verghese, G. (2023, March 13). Understand and trust data with Dataplex data lineage. *Google Cloud Blog*.

How to use the Dataplex data lineage feature

- Google Cloud. (2023, November 29). *About data lineage*.
- Google Cloud. (2024, January 2). *Track data lineage for a BigQuery table*.

Introduction to Analytics Hub

- Alfonso, A. (2022, April 25). Google Cloud—Analytics Hub. *Medium*.
- Google Cloud. (2023, November 29). *Introduction to Analytics Hub*.
- Google Cloud Tech. (2021, October 18). *Analytics Hub in a minute* [Video]. YouTube.

Analytics Hub enables data sharing

- Alfonso, A. (2022, April 25). Google Cloud—Analytics Hub. *Medium*.

- Google Cloud. (n.d.). *Analytics hub*. Retrieved September 25, 2023.
- Google Cloud. (2024, January 2). *Configure Analytics Hub roles*.
- Google Cloud Tech. (2021, October 18). *Analytics Hub in a minute* [Video]. YouTube.

How to use Analytics Hub

- Alfonso, A. (2022, April 25). Google Cloud—Analytics Hub. *Medium*.
- Google Cloud. (2024, January 2). *Manage data exchanges*.

Data discovery, curation, and unification

- Google Cloud. (n.d.). *Dataplex*. Retrieved December 28, 2023.
- Google Cloud. (2023, December 20). *Discover data*. Retrieved December 28, 2023.
- Google Cloud Tech. (2020, November 5). *Google Cloud Data Catalog essentials: Data discovery* [Video]. YouTube.

Overview of Dataplex

- Damle, P., & Kothari, N. (2022, February 22). Build a data mesh on Google Cloud with Dataplex, now generally available. *Google Cloud Blog*.
- Google Cloud. (2023, December 4). *Dataplex overview*. Retrieved December 5, 2023.
- Google Cloud Events. (2022, April 6). *Build a data mesh on GCP with Dataplex* [Video]. YouTube.
- Google Cloud Events. (2022, July 27). *Manage and govern distributed data with Dataplex* [Video]. YouTube.
- Lenka, B. K. (2023, July 3). *Data mesh vs data fabric architectures*. Linkedin.

Benefits of using Dataplex

- Google Cloud. (n.d.). *Dataplex*. Retrieved December 5, 2023.

- Google Cloud. (2023, December 4). *Dataplex overview*. Retrieved December 5, 2023.
- Seabrook, E. (2023, May 10). *Dataplex: Data mesh and governance made easy*. Linkedin.

How to search for data with BigQuery

- Google Cloud. (2024, January 8). *Introduction to search in BigQuery*.
- Google Cloud. (2024, January 8). *Search functions*.

Module 4: Techniques to access data

Resources

Auto-detection of schemas in BigQuery

- Using schema auto-detection

Cloud-based machine learning can train predictive models

- Introduction to Vertex AI

Managed notebooks

- Vertex AI Workbench Jupyter Notebook tutorials
- Welcome to Colab!

Benefits of data partitioning

- Database partitions: How they work

Data partitioning reduces cloud costs

- Cloud Storage pricing
- Understanding Data Partitioning in Data Engineering: A Key to Scalability and Performance

Create a partitioned table

- Creating partitioned tables

Tips for interacting with partitioned tables

- Load data into partitioned tables
- Managing partitioned tables

How to manage Dataproc clusters

- Create and manage Labels
- Stop and start Dataproc clusters

Interview tip: Provide examples

- Interview Warmup

Citations

Methods for defining BigQuery table schemas

- Bapna, H. (2019, May 30). Bigquery: Table schema. *Medium*.
- Google Cloud. (2023, December 4). *Loading JSON data from Cloud Storage*.
- Google Cloud. (2023, December 4). *Specify nested and repeated columns in table schemas*.
- Google Cloud. (2023, December 4). *Specifying a schema*.

Auto-detection of schemas in BigQuery

- Google Cloud. (2024, January 2). *Modifying table schemas*.
- Google Cloud. (2024, January 2). *Using schema auto-detection*.

Basic SQL commands for querying data

- Indiana University. (2023, July 7). *SQL example statements for retrieving data from a table*.
- Maryville University. (2020, August 5). *SQL guide for beginner data scientists and developers*.
- MySQL. (n.d.). *Chapter 7 Examples of Common Queries*.

[Supplemental] SQL query terms

- W3 schools. (n.d.). *SQL keywords reference*.

Steps and models for accessing data with machine learning

- Google Cloud. (n.d.). *Classification and regression overview*. Retrieved December 5, 2023.
- Google Cloud. (n.d.). *What is machine learning (ML)?* Retrieved December 5, 2023.
- Google Cloud. (2022, December 15). *Best practices for implementing machine learning on Google Cloud*. Retrieved December 5, 2023.
- Google Cloud. (2023, December 18). *Google Cloud APIs*.
- Google Cloud Tech. (2022, August 5). *Machine learning on Google Cloud* [Video]. YouTube.
- Google for Developers. (2022, July 18). *What is clustering?* Retrieved December 5, 2023.

Cloud-based machine learning can train predictive models

- Google Cloud. (n.d.). *What is machine learning (ML)?* Retrieved December 5, 2023.
- Google Cloud. (2024, January 2). *Introduction to Vertex AI*.

Introduction to machine learning with Vertex AI and BigQuery

- Caram, F. (2022, April 20). Machine learning and artificial intelligence in GCP. *Avenue Code Snippets*.
- Google Cloud. (n.d.). *Bigquery*. Retrieved December 5, 2023.
- Google Cloud. (n.d.). *Vertex AI*. Retrieved December 5, 2023.
- Narayana, V., & Geerdink, B. (2022, June 16). Wayfair: Accelerating MLOps to power great experiences at scale. *Google Cloud Blog*.

Overview of Google Colab

- Google. (n.d.). *Colaboratory*. Retrieved June 8, 2023.
- Google Colaboratory. (n.d.). *Welcome to Colab!* Colab Sandbox. Retrieved June 8, 2023.
- Jupyter. (n.d.). *The Jupyter Notebook*.
- Jupyter. (n.d.). *User interface components*.
- Python. (n.d.). *What is Python? Executive summary*. Retrieved June 8, 2023.

Managed notebooks

- Google Cloud. (2024, January 2). *Introduction to Vertex AI Workbench*.
- Google Cloud. (2024, January 2). *Vertex AI Workbench Jupyter Notebook tutorials*.
- Google Cloud Tech. (2023, March 29). *Colab and Vertex AI Workbench #Jupyter notebooks #shorts* [Video]. YouTube.
- Google Colaboratory. (n.d.). *Welcome to Colab!* Colab Sandbox. Retrieved June 8, 2023.

Essentials of database partitioning

- Namdeo, S. (2021, February 5). Data partitioning—System design concept. *Medium*.

Benefits of data partitioning

- Nestler, J. (2021, May 28). *Database partitions: How they work*. Looker.
- Nukala, Y. R. (2023, May 16). *Understanding data partitioning in data engineering: A key to scalability and performance*. Linkedin.

Methods for partitioning tables

- Google Cloud. (2023, December 20). *Introduction to partitioned tables*.
- Google Cloud. (2024, January 2). *Introduction to clustered tables*.

Data partitioning reduces cloud costs

- Google Cloud. (n.d.). *Cloud Storage pricing*. Retrieved January 2, 2024.
- Nukala, Y. R. (2023, May 16). *Understanding data partitioning in data engineering: A key to scalability and performance*. Linkedin.

Create a partitioned table

- Google Cloud. (2024, January 2). *Creating partitioned tables*.

Strategies for querying partitioned tables

- Google Cloud. (2023, December 20). *Query partitioned tables*.

Tips for interacting with partitioned tables

- Google Cloud. (2024, January 2). *Export table data to Cloud Storage*.
- Google Cloud. (2024, January 2). *Introduction to partitioned tables*.
- Google Cloud. (2024, January 2). *Updating partitioned table data using DML*.

Key processes and benefits of Dataproc

- Apache Spark. (2023, September 9). *PySpark overview*.
- Google Cloud. (n.d.). *What is Apache Hadoop?*
- Google Cloud. (2023, December 21). *Create a Dataproc cluster by using the Google Cloud console*.
- Google Cloud. (2023, December 21). *What is Dataproc?*

How to create a Dataproc cluster

- Google Cloud. (2023, December 21). *Create a Dataproc cluster by using the Google Cloud console*.

How to manage Dataproc clusters

- Google Cloud. (2024, January 4). *Create and manage labels*.
- Google Cloud. (2024, January 4). *Manage a cluster*.
- Google Cloud. (2024, January 4). *Stop and start Dataproc clusters*.